Machine Learning Unit 3: Concept Learning & Candidate Elimination Algorithm

1. Candidate Elimination Algorithm:
The Candidate Elimination algorithm computes the version space containing all hypotheses from the hypothesis space H that are consistent with an observed sequence of training examples. It initializes the Specific boundary S to the most specific hypothesis and the General boundary G to the most general hypothesis.

2. Version Space Representation:
A hypothesis h is consistent with a set of training examples D if and only if h(x) = c(x) for each example (x, c(x)) in D. As new positive examples arrive, S is generalized; as negative examples arrive, G is specialized.

3. Key Takeaway:
Candidate elimination efficiently prunes hypotheses without enumerating all possible concepts.